



SHORE

Year 12
Mathematics - Extension 2
Trial Examination
2010

Student Number: _____

Maths set: 12ME2-1

General Instructions

- * Reading time – 5 minutes
- * Working time – 3 hours
- * Write using black or blue pen
- * Board-approved calculators may be used
- * All necessary working should be shown in every question
- * A table of standard integrals is attached on the final page

Note: Any time you have remaining should be spent revising your answers.

Total marks - 120

- * Attempt Questions 1 - 8
- * All questions are of equal value
- * Start each question in a new writing booklet
- * Write your examination number on the front cover of each booklet to be handed in
- * If you do not attempt a question, submit a blank booklet with your examination number and "N/A" on the front cover

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

Question 1 (15 Marks)

Use a Separate Booklet

Marks

a) Evaluate $\int_{\frac{\pi}{16}}^{\frac{\pi}{12}} \sec 4x \tan 4x \, dx$

2

b) Find $\int x \ln x \, dx$

2

c) Find $\int \frac{9x^3 + 9x^2 + 5x + 4}{3x + 1} \, dx$

3

d) i. Find constants a , b and c such that

2

$$\frac{3x^2 - 2x - 3}{(x^2 + 9)(x - 3)} = \frac{ax + b}{x^2 + 9} + \frac{c}{x - 3}$$

ii. Hence find $\int \frac{3x^2 - 2x - 3}{(x^2 + 9)(x - 3)} \, dx$.

2

e) By making the substitution $t = \tan \frac{\theta}{2}$, evaluate $\int_0^{\frac{\pi}{2}} \frac{d\theta}{1 + \sin \theta + \cos \theta}$

4

End of Question 1

Question 2 (15 Marks)

Use a Separate Booklet

Marks

- a) Given $A = 3 + 4i$ and $B = 1 - i$, express the following in the form $x + iy$ where x and y are real numbers.

i. AB

1

ii. $\frac{A}{iB}$

2

iii. $\sqrt{A}$ (give both possible answers)

3

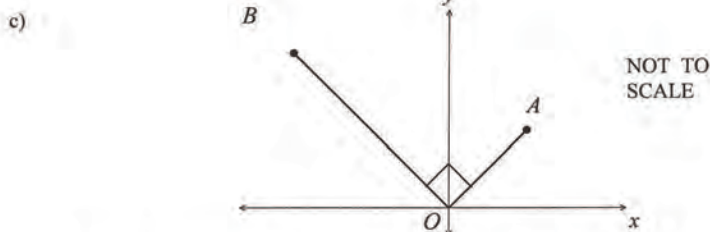
- b) If $w = \sqrt{3} - i$,

i. Find the exact value of $|w|$ and $\arg w$.

2

ii. Find the exact value of w^3 in the form $a + ib$ where a and b are real.

2



On the Argand diagram, OA represents the complex number $z_1 = x + iy$, $\angle AOB = \frac{\pi}{2}$ and the length of OB is twice that of OA.

i. Show that OB represents the complex number $-2y + 2ix$.

1

ii. Given that AOCB is a rectangle, find the complex number represented by OC.

1

iii. Find the complex number represented by BA.

1

- d) Sketch the region on an argand diagram where

2

$$|z - 1| \leq \sqrt{2} \text{ and } 0 \leq \arg(z + i) \leq \frac{\pi}{4} \text{ both hold.}$$

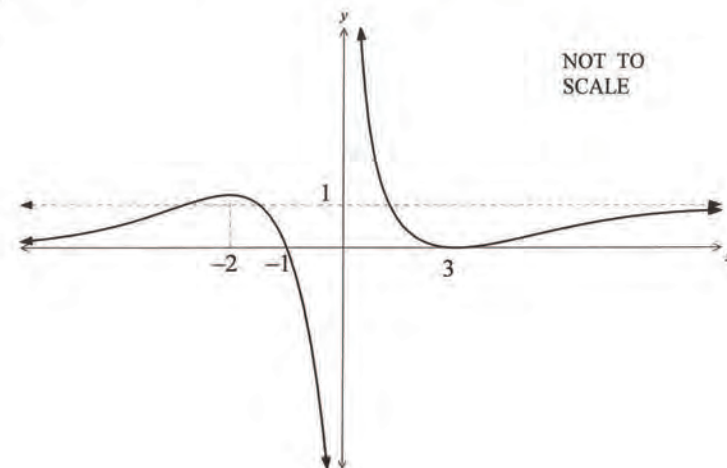
End of Question 2

Question 3 (15 Marks)

Use a Separate Booklet

Marks

- a)



The diagram shows the graph of the function $y = f(x)$ which has asymptotes, vertically at $x = 0$ and horizontally at $y = 1$ for $x \geq 0$ and at $y = 0$ for $x \leq 0$.

Draw separate sketches of the following showing any critical features.

i. $y = \frac{1}{f(x)}$

3

ii. $y = [f(x)]^2$

3

iii. $y = f'(x)$

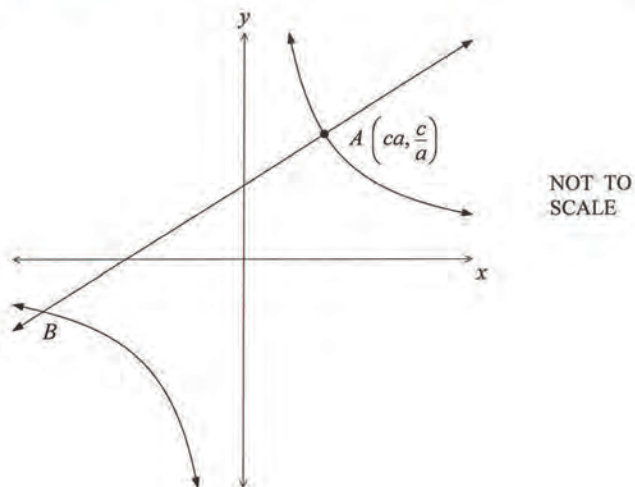
2

Question 3 continues

Question 3 continued

Marks

b)



The point $A\left(ca, \frac{c}{a}\right)$, where $a \neq \pm 1$ lies on the hyperbola $xy = c^2$. The normal through A meets the other branch of the curve at B .

- i. Show that the equation of the normal through A is

2

$$y = a^2x + \frac{c}{a}(1 - a^4)$$

- ii. Hence if B has coordinates $\left(cb, \frac{c}{b}\right)$, show that $b = \frac{-1}{a^3}$.

3

- iii. If this hyperbola is rotated clockwise through 45° , show that the equation becomes

2

$$X^2 - Y^2 = 2c^2.$$

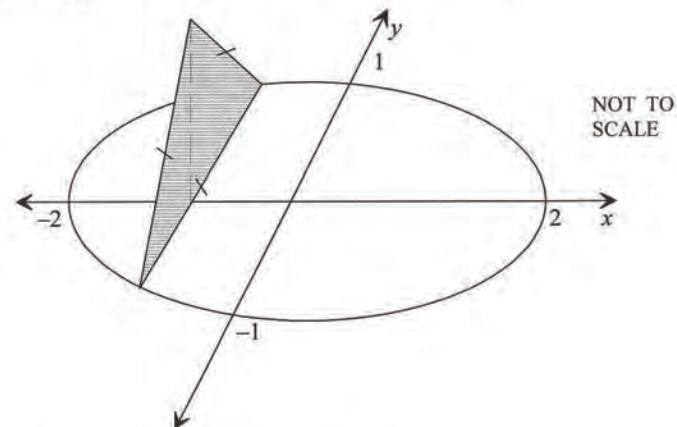
End of Question 3

Question 4 (15 Marks)

Use a Separate Booklet

Marks

- a) A solid shape has as its base an ellipse in the XY plane as shown below. Sections taken perpendicular to the X -axis are equilateral triangles. The major and minor axes of the ellipse are 4 metres and 2 metres respectively.



- i. Write down the equation of the ellipse.

1

- ii. Show that the area of the cross-section is given by

2

$$A = \frac{\sqrt{3}}{4}(4 - x^2).$$

- iii. By using the technique of slicing, find the volume of the solid.

2

- b) The region enclosed by the curve $y = 5x - x^2$, the x axis and the lines $x = 1$ and $x = 3$ is rotated about the y axis. By using the method of cylindrical shells, find the volume of the solid so produced.

4

Question 4 continues

Question 4 continued

Marks

- c) The roots of the equation $x^3 - 3x^2 + 9 = 0$ are α , β and γ .
- Determine the polynomial equation with roots α^2 , β^2 and γ^2 .
 - Find the value of $\alpha^2 + \beta^2 + \gamma^2$ and hence evaluate $\alpha^3 + \beta^3 + \gamma^3$.
- d) Given that the polynomial $P(x)$ has a double root at $x = \alpha$, show that the polynomial $P'(x)$ will have a single root at $x = \alpha$.

2

2

2

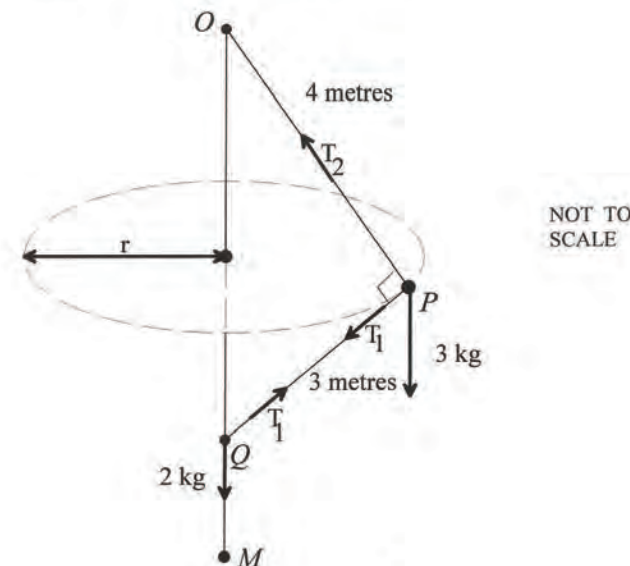
End of Question 4

Question 5 (15 Marks)

Use a Separate Booklet

Marks

a)



The above sketch shows a smooth vertical rod OM . Light inextensible strings OP and QP are attached to the rod at O and a mass of 3 kg at P . At Q , a 2 kg mass is free to slide on the rod. P is rotating in a horizontal circle about the rod and maintains a right angle at P . Let angle POQ be θ .

Note that the distance OQ is 5 metres .

- By considering the forces vertically at P and Q calculate the tension T_1 in PQ and T_2 in OP . (In terms of g)
- By considering the forces horizontally at P calculate the angular velocity of P to maintain this system. Give your answer correct to one decimal place. (Use $g = 10\text{ ms}^{-2}$)

4

3

Question 5 continues

Question 5 continued

- | | Marks |
|---|-------|
| b) By taking logarithms of both sides and then differentiating implicitly, verify the rule for differentiating the quotient $y = \frac{u(x)}{v(x)}$ is given by | 3 |
| $\frac{dy}{dx} = \frac{v(x)u'(x) - u(x)v'(x)}{(v(x))^2}$ | |
| c) i. Show that the recurrence (reduction) formula for | 3 |
| $I_n = \int \tan^n x dx \quad \text{is} \quad I_n = \frac{1}{n-1} \tan^{n-1} x - I_{n-2}.$ | |
| ii. Hence evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x dx$ | 2 |

End of Question 5

Question 6 (15 Marks)

Use a Separate Booklet

Marks

- | | |
|--|---|
| a) A solid of unit mass is dropped from above the ground. Air resistance is proportional to the speed (v) of the mass. (acceleration under gravity = g) | |
| i. Write the equation for the acceleration of the mass. (Use k as the constant of proportionality) | 1 |
| ii. Show that the velocity (v) of the solid after t seconds is given by | 3 |
| $v = \frac{g}{k} (1 - e^{-kt})$ | |
| iii. Show that the displacement (x) of the solid at velocity v is given by | 3 |
| $x = \frac{g}{k^2} \left[\ln \frac{g}{g - kv} - \frac{kv}{g} \right].$ | |
| iv. Using either part (ii) or (iii) deduce the terminal velocity of the mass. | 1 |
| | |
| b) Given $z = \cos \theta + i \sin \theta$, and using De Moivre's Theorem | |
| i. Find an expression for $\cos 4\theta$ in terms of powers of $\cos \theta$. | 3 |
| ii. Determine the roots of the equation $z^4 = -1$. | 2 |
| iii. Using the fact that $z^n + \frac{1}{z^n} = 2 \cos n\theta$, find an expression for $\cos^4 \theta$ in terms of $\cos 2\theta$ and $\cos 4\theta$. | 2 |

End of Question 6

Question 7 (15 Marks)

Use a Separate Booklet

Marks

- a) When a polynomial $P(x)$ is divided by $(x-1)$ the remainder is 3 and when divided by $(x-2)$ the remainder is 5. Find the remainder when the polynomial is divided by $(x-1)(x-2)$. 2
- b) i. Prove that $\cos[(k-1)\theta] - 2\cos\theta\cos k\theta = -\cos[(k+1)\theta]$. 1
- ii. Hence, using mathematical induction, prove that if n is a positive integer then 4
- $$1 + \cos\theta + \cos 2\theta + \dots + \cos(n-1)\theta = \frac{1 - \cos\theta - \cos n\theta + \cos[(n-1)\theta]}{2 - 2\cos\theta}$$
- c) A mass of 20 kg hangs from the end of a rope under gravity and is hauled up vertically from rest by winding up the rope. The pulling force on the rope starts at 250 N and decreases uniformly by 10 N for every metre wound up. In other words, the pulling force is $250 - 10x$ Newtons where x m is the height above the initial starting point. 3
- Find the velocity of the mass when 10 metres have been wound up.
- (Neglect the weight of the rope and take $g = 10 \text{ ms}^{-2}$)
- d) i. Show that $\frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta} = \sin\theta + i\cos\theta$ where $\theta \neq \frac{3\pi}{2} + 2k\pi$; k integer. 3
- ii. Hence prove that $\left(\frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta}\right)^n = \cos\left(\frac{n\pi}{2} - n\theta\right) + i\sin\left(\frac{n\pi}{2} - n\theta\right)$ 2
- where n is a positive integer and where $\theta \neq \frac{3\pi}{2} + 2k\pi$; k integer.

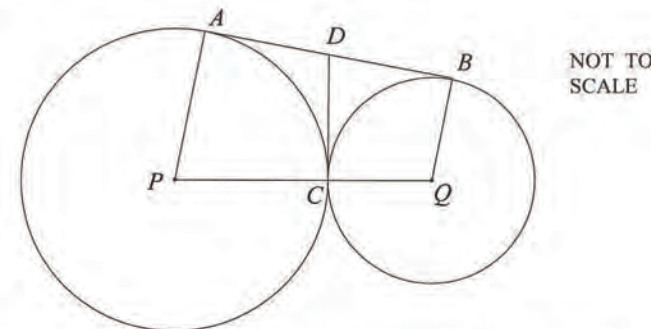
End of Question 7

Question 8 (15 Marks)

Use a Separate Booklet

Marks

a)



In the diagram PCQ is a straight line joining the centres of the circles P and Q. AB and DC are common tangents.

Copy or trace the diagram into your writing booklet.

- i. Explain why CDBQ is a cyclic quadrilateral. 2
- ii. Show that $\triangle ADC \parallel \triangle BQC$. 3
- iii. Show that $PD \parallel CB$. 3
- b) Given $2\cos A \sin B = \sin(A+B) - \sin(A-B)$
- If $P = 1 + 2\cos\theta + 2\cos 2\theta + 2\cos 3\theta$
- i. Prove that $P \sin \frac{\theta}{2} = \sin \frac{7\theta}{2}$. 2
- ii. Hence show that $1 + 2\cos \frac{2\pi}{7} + 2\cos \frac{4\pi}{7} + 2\cos \frac{6\pi}{7} = 0$ 2
- iii. By writing P in terms of $\cos\theta$, prove that $\cos \frac{2\pi}{7}$ is a root of the polynomial equation 3

$$8x^3 + 4x^2 - 4x - 1 = 0$$

End of Examination

Solution: Math 25 Est 2 Trial HSC 2010 Shore

$$1(a) I = \int_{\pi/16}^{\pi/12} \sec 4x \, dx$$

$$= \frac{1}{4} (\sec \frac{\pi}{3} - \sec \frac{\pi}{4})$$

$$= \frac{1}{4} (2 - \sqrt{2})$$

$$\text{OR } \frac{2 - \sqrt{2}}{4} \quad (2)$$

$$(b) I = \int \ln x \, d(\frac{x}{2})$$

$$= \frac{x}{2} \ln x - \int \frac{x}{2} d(\ln x)$$

$$= \frac{x}{2} \ln x - \int \frac{1}{2} dx$$

$$= \frac{x}{2} \ln x - \frac{1}{2} x + c$$

$$\text{OR } \frac{1}{2} x^2 (2 \ln x - 1) + c \quad (2)$$

$$(c) 3x + \sqrt{9x^2 + 9x^2 + 5x + 4}$$

$$\frac{6x^2 + 5x}{6x^2 + 5x}$$

$$\frac{3x + 4}{2x + 1}$$

$$I = \int 3x^2 + 2x + 1 + \frac{3}{2x+1} dx$$

$$= x^3 + x^2 + x + \ln|2x+1| + c \quad (3)$$

$$(d)(i) \text{ If } k=3, \frac{27-6-3}{18} = c$$

$$\therefore c = 1$$

$$\text{Also } 3x^2 - 2x - 3 = (ax+b)(x-3) + c(x^2+9)$$

$$\text{Equate coeff } x: a + c = 3$$

$$\text{Equate const: } -3 = -3b + 9c$$

$$\therefore a = 2 \text{ and } b = 4 \quad (2)$$

$$(ii) I = \int \frac{2x+4}{x^2+9} + \frac{1}{2-3} dx$$

$$= \int \frac{2x}{x^2+9} + \frac{4}{x^2+3} + \frac{1}{x-3} dx$$

$$= \ln(x^2+9) + \frac{4}{3} \tan^{-1} \frac{x}{\sqrt{3}} + \ln|x-3| + c$$

$$\text{OR } \ln(x^2+9)(x-3) + \frac{4}{3} \tan^{-1} \frac{x}{\sqrt{3}} + c \quad (2)$$

$$(b) t = \tan \theta \Rightarrow \theta = 2 \tan^{-1} t \Rightarrow \frac{d\theta}{dt} = \frac{2}{1+t^2}$$

$$\therefore d\theta = \frac{2dt}{1+t^2} : \theta = 0, t = 0; \theta = \frac{\pi}{2}, t = 1$$

$$\text{Also } \tan \theta = \frac{2t}{1-t^2}, \sin \theta = \frac{2t}{1+t^2}, \cos \theta = \frac{1-t^2}{1+t^2}$$

$$\therefore I = \int_0^1 \frac{2dt}{1+t^2} + \frac{1-t^2}{1+t^2} \times \frac{1}{1+t^2}$$

$$\therefore I = \int_0^1 \frac{2dt}{1+t^2+2t+1-t^2}$$

$$= \int_0^1 \frac{dt}{1+t}$$

$$= [\ln|1+t|]_0^1$$

$$= \ln 2 - \ln 1$$

$$= \ln 2 \quad (4)$$

$$2(a)(i) AB = (3+4i)(1-i)$$

$$= 3+4-3i+4i$$

$$= 7+i \quad (1)$$

$$(ii) \frac{A}{iB} = \frac{3+4i}{i+1} \times \frac{i-1}{i-1}$$

$$= \frac{-3-4-4i+3i}{-1-1}$$

$$= \frac{-7-i}{-2} = \frac{7}{2} + \frac{i}{2} \quad (2)$$

$$(iii) \sqrt{A} = \sqrt{3+4i} = x+iy \text{ say}$$

$$\therefore 3+4i = x^2-y^2+2xyi$$

$$\text{Equate: } x^2-y^2=3 \text{ \& } xy=2$$

$$\therefore x=2, y=\pm 1$$

$$\therefore \sqrt{3+4i} = 2+i \text{ or } 2-i \quad (3)$$

$$(b)(i) \text{ real } |w| = 2, \arg w = \frac{\pi}{6} \quad (2)$$

$$(ii) w^5 = [2 \cos(\frac{\pi}{6})]^5$$

$$= 2^5 \cos \frac{5\pi}{6}$$

$$= 32(-\frac{\sqrt{3}}{2} - \frac{i}{2})$$

$$= -16\sqrt{3} - 16i \quad (2)$$

$$(c)(i) OB \text{ represents } 2i(x+iy) = 2ix - 2y$$

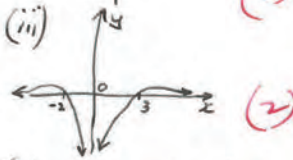
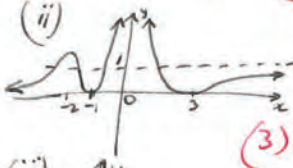
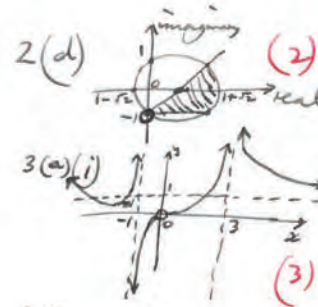
$$= -2y + 2ix \quad (1)$$

$$(ii) OC \text{ represents } (x+iy) + (-2y+2ix) = (x-2y) + 2ix$$

$$(ii) \vec{BA} = \vec{BO} + \vec{OA}$$

$$= (2y-2ix) + (x+iy)$$

$$= (x+2y) + i(-2x+y) \quad (1)$$



$$(b) \text{ Now } x = ca \text{ and } y = \frac{c}{a}$$

$$(i) \therefore \frac{dx}{da} = c \text{ \& } \frac{dy}{da} = -\frac{c}{a^2}$$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{da}}{\frac{dx}{da}} = \frac{-\frac{c}{a^2}}{c} = -\frac{1}{a^2}$$

$$= -\frac{1}{x^2}$$

$$\text{Normal grad } a^2$$

$$\therefore \frac{y-cx}{x-\frac{c}{a}} = a^2$$

$$\therefore y-cx = a^2x - a^2c$$

$$\therefore y = a^2x + \frac{c}{a} - a^2c$$

$$= a^2c + \frac{c}{a}(1-a^4) \quad (2)$$

$$(ii) \text{ Sub in } (cb, \frac{c}{b})$$

$$\therefore \frac{c}{b} = a^2cb + \frac{c}{a}(1-a^4)$$

$$(x \neq 0): a = a^3b^2 + b - ba^4$$

$$\therefore 0 = a^3b^2 + b - ba^4 - a$$

$$= b(a^3b+1) - a(ba^3+1)$$

$$= (a^3b+1)(b-a)$$

$$\text{But } a \neq \pm 1 \therefore a \neq b \text{ by symmetry}$$

$$\therefore a^3b = -1$$

$$\therefore b = -\frac{1}{a^3} \quad (3)$$

(iii) Through 45° it becomes:



This is of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$
But it is rectangular $\therefore a=b$
 $\therefore$ form $x^2 - y^2 = k^2$ for some k .
On original hyperbola (5c) is $\sqrt{2}c$ from the origin
 $\therefore OP = \sqrt{2}c \therefore P(\sqrt{2}c, 0)$
Sub in $(\sqrt{2}c)^2 - 0^2 = k^2 \Rightarrow k = \sqrt{2}c$
 $\therefore$ new eqn $x^2 - y^2 = 2c^2 \quad (2)$

$$4(a)(i) \frac{x^2}{4} + y^2 = 1 \quad (1)$$

$$(ii) y^2 = 1 - \frac{x^2}{4}$$

$$\therefore 2y = 2\sqrt{1-\frac{x^2}{4}}, \text{ each side of the equilateral } \Delta$$

$$\text{Area} = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} \times 2\sqrt{1-\frac{x^2}{4}} \times 2\sqrt{1-\frac{x^2}{4}} \sin 60^\circ$$

$$= \sqrt{3}(1-\frac{x^2}{4})$$

$$= \frac{\sqrt{3}}{4}(4-x^2) \quad (2)$$

$$(iii) \Delta V = \frac{\sqrt{3}}{4}(4-x^2) dx$$

$$\therefore V = \lim_{\Delta k \rightarrow 0} \sum_{k=-2}^2 \frac{\sqrt{3}}{4}(4-k^2) \Delta k$$

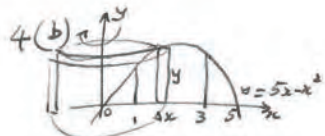
$$= \int_{-2}^2 \frac{\sqrt{3}}{4}(4-k^2) dk$$

$$= 2 \int_0^2 \frac{\sqrt{3}}{4}(4-k^2) dk$$

$$= 2 \times \frac{\sqrt{3}}{4} [4k - \frac{k^3}{3}]_0^2$$

$$= \frac{\sqrt{3}}{2} (8 - \frac{8}{3})$$

$$= \frac{8\sqrt{3}}{3} u^3 \quad (2)$$



Ignoring 2nd order differences

$$\delta V \approx 2\pi x \cdot y \cdot \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum 2\pi xy \delta x$$

$$= \int_0^3 2\pi x (5x - x^2) dx$$

$$= \int_0^3 2\pi (5x^2 - x^3) dx$$

$$= 2\pi \left[\frac{5x^3}{3} - \frac{x^4}{4} \right]_0^3$$

$$= 2\pi \left(45 - \frac{81}{4} - \frac{5}{3} + \frac{1}{4} \right)$$

$$= \frac{160\pi}{3} \text{ m}^3 \quad (4)$$

(c) $x = t, y = t^2, z = t^3$

(i) $\therefore y = x^2 \Rightarrow x = \sqrt{y}$ say

$$\therefore (\sqrt{y})^3 - 3(\sqrt{y})^2 + 9 = 0$$

$$\therefore y\sqrt{y} - 3y + 9 = 0$$

$$\therefore y\sqrt{y} = 3y - 9$$

$$\therefore y^2 = (3y - 9)^2$$

$$\therefore y^2 = 9y^2 - 54y + 81$$

$$\therefore \underline{y^2 - 9y^2 + 54y - 81 = 0} \quad (2)$$

(ii) Using sum of the roots

$$x^2 + y^2 + z^2 = 9$$

$$\text{Now } x^3 - 3x^2 + 9 = 0$$

$$y^3 - 3y^2 + 9 = 0$$

$$z^3 - 3z^2 + 9 = 0$$

$$\therefore x^3 + y^3 + z^3 - 3(x^2 + y^2 + z^2) + 27 = 0$$

$$\therefore x^3 + y^3 + z^3 = 27 - 27$$

$$= 0 \quad (2)$$

(d) Let $P(x) = (x-1)^2 \cdot Q(x)$ where $Q(x) \neq 0$

$$\therefore P'(x) = Q(x) \cdot 2(x-1)' + (x-1)^2 \cdot Q'(x)$$

$$= (x-1) [2Q(x) + (x-1) \cdot Q'(x)]$$

$$\therefore P'(x) = 0 \Rightarrow P'(x) \text{ has a single root as}$$

$Q(x) \neq 0$ with $(x-1)$ not being a factor.

5 (c) Horizontally at P: (2)

$$m r \omega^2 = T_1 \sin \theta + T_2 \sin \theta$$

$$\therefore 3 r \omega^2 = T_1 \sin \theta + T_2 \sin \theta$$

$$\text{Now } \frac{r}{4} = \sin \theta, \text{ but } \sin \theta = \frac{3}{5}$$

$$\therefore r = \frac{12}{5}, \sin \theta = \frac{12}{13}$$

$$\therefore 3 \left(\frac{12}{5} \right) \omega^2 = T_1 \cdot \frac{4}{5} + T_2 \cdot \frac{3}{5}$$

$$\therefore 36 \omega^2 = 4T_1 + 3T_2 \quad (1)$$

Vertically at Q: $0 = T_1 \cos \theta - 2g$

$$= T_1 \cdot \frac{3}{5} - 2g$$

$$\therefore T_1 = \frac{10g}{3} \text{ N}$$

Vertically at P: $0 = T_2 \cos \theta - T_1 \cos \theta - 3g$

$$= T_2 \cdot \frac{4}{5} - \frac{10g}{3} \cdot \frac{3}{5} - 3g$$

$$\therefore T_2 \cdot \frac{4}{5} = 9g$$

$$\therefore T_2 = \frac{25g}{4} \text{ N} \quad (4)$$

Sub into (1) $36 \omega^2 = 4 \times \frac{10g}{3} + 3 \times \frac{25g}{4}$

$$\therefore \omega^2 = \frac{1925}{216} \text{ where } g = 10$$

$$\therefore \omega \approx 3.0 \text{ rad.s}^{-1} \quad (3)$$

(b) $y = \frac{x}{2}$

$$\therefore \ln y = \ln x - \ln 2$$

$$\therefore \frac{1}{y} \times \frac{dy}{dx} = \frac{1}{x} \times \frac{dx}{dx} - \frac{1}{x} \times \frac{dx}{dx}$$

$$\therefore \frac{1}{x} \times \frac{dy}{dx} = \frac{1}{x} \times \frac{dx}{dx} - \frac{1}{x} \times \frac{dx}{dx}$$

$$\therefore \frac{dy}{dx} = \frac{1}{x} \times \frac{dx}{dx} \times \frac{x}{2} - \frac{1}{x} \times \frac{dx}{dx} \times \frac{x}{2}$$

$$= \frac{1}{2} \times \frac{dx}{dx} - \frac{1}{2} \times \frac{dx}{dx}$$

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$$5 (c) (i) I_n = \int \tan^{n-2} x \cdot \tan^2 x dx$$

$$= \int \tan^{n-2} x (\sec^2 x - 1) dx$$

$$= \int \tan^{n-2} x \sec^2 x dx - I_{n-2}$$

$$\text{Let } u = \tan x, du = \sec^2 x dx$$

$$\therefore I_n = \int u^{n-2} du - I_{n-2}$$

$$= \frac{u^{n-1}}{n-1} - I_{n-2}$$

$$= \frac{\tan^{n-1} x}{n-1} - I_{n-2} \quad (3)$$

$$\therefore I_3 = \left[\frac{\tan^2 x}{2} \right]_0^{\pi/4} - \int_0^{\pi/4} \tan x dx$$

$$= \frac{1}{2} - \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$$

$$\text{Let } v = \cos x, dv = -\sin x dx$$

$$\therefore I_3 = \frac{1}{2} + \int_1^{\frac{1}{\sqrt{2}}} \frac{1}{v} dv$$

$$= \frac{1}{2} + \left[\ln v \right]_1^{\frac{1}{\sqrt{2}}}$$

$$= \frac{1}{2} + \ln \frac{1}{\sqrt{2}} - \ln 1$$

$$= \frac{1}{2} - \frac{1}{2} \ln 2$$

$$= \frac{1}{2} (1 - \ln 2) \quad (2)$$

6. $T = 0, x = H$

(i) Here $m = 1$ & $F = ma$

$$\therefore ma = g - kv \quad (1)$$

(ii) $\therefore \frac{dv}{dt} = g - kv$

$$\therefore \frac{dv}{g - kv} = dt$$

$$\therefore \int \frac{-kv}{g - kv} = \int -k dt$$

$$\therefore \left[\ln |g - kv| \right]_0^v = \left[-kt \right]_0^t$$

$$\therefore \ln |g - kv| - \ln g = -kt$$

$$\therefore -kt = \ln \left| \frac{g - kv}{g} \right|$$

$$\therefore -kt = \ln |g - kv| - \ln g$$

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$$(iii) v \frac{dv}{dx} = g - kv$$

$$\therefore \frac{v dv}{g - kv} = dx$$

$$\frac{1}{g - kv} \cdot \frac{v dv}{dx} = \frac{1}{g - kv}$$

$$\therefore \left(-\frac{1}{k} + \frac{g}{g - kv} \right) dv = dx$$

$$\therefore \int \left(-\frac{1}{k} + \frac{g}{g - kv} \right) dv = \int dx$$

$$\therefore \left[-\frac{v}{k} - \frac{g}{k^2} \ln |g - kv| \right]_0^v = \left[x \right]_0^x$$

$$\therefore -\frac{v}{k} - \frac{g}{k^2} \ln |g - kv| + \frac{g}{k^2} \ln g = x - 0$$

$$\therefore -\frac{v}{k} - \frac{g}{k^2} \ln |g - kv| + \frac{g}{k^2} \ln g = x - 0$$

$$\therefore x = \frac{g}{k^2} \left[-\frac{kv}{g} + \frac{1}{g - kv} \right] \quad (3)$$

(iv) From part (ii) $\lim_{T \rightarrow \infty} (e^{-kT}) = 0$

$$\therefore \text{Terminal velocity } W = \frac{g}{k} \quad (1)$$

(b) (i) $(\cos 4\theta) = (\cos \theta)^4$ by de Moivre's Rule

$$\therefore \cos 4\theta + i \sin 4\theta = \cos^4 \theta + 4i \cos^3 \theta \sin \theta + 6 \cos^2 \theta \sin^2 \theta$$

$$- 4i \cos \theta \sin^3 \theta + \sin^4 \theta$$

$$\text{Equating } \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

$$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2$$

$$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta$$

$$= 8 \cos^4 \theta - 8 \cos^2 \theta + 1 \quad (4) \quad (3)$$

$$(ii) \cos 4\theta = \cos \pi \text{ or } \cos 3\pi \text{ or } \cos 5\pi \text{ or } \cos 7\pi$$

$$\therefore 4\theta = \pi, 3\pi, 5\pi, 7\pi$$

$$\therefore \theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4} \text{ or } \frac{5\pi}{4} \text{ or } \frac{7\pi}{4}$$

$$\therefore z = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \text{ or } \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \text{ or } \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \text{ or } \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4}$$

$$\therefore z = \cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \text{ or } \cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \text{ or } \cos \frac{5\pi}{4} + i \sin \frac{5\pi}{4} \text{ or } \cos \frac{7\pi}{4} + i \sin \frac{7\pi}{4} \quad (2)$$

$$(iii) \text{ Let } n = 1 \therefore z + \frac{1}{z} = 2 \cos \theta$$

$$\therefore \left(z + \frac{1}{z} \right)^4 = (2 \cos \theta)^4$$

$$\therefore \cos^4 \theta = \frac{1}{16} \left(z^4 + 4z^2 + 6 + 4\frac{1}{z^2} + \frac{1}{z^4} \right)$$

$$= \frac{1}{16} (2 \cos 4\theta + 4 \times 2 \cos 2\theta + 6)$$

$$= \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8} \quad (2)$$

$$\text{OR } 8 \cos^4 \theta = \cos 4\theta + 8 \cos^2 \theta + 1 \text{ from (4)}$$

$$= \cos 4\theta + 4(2 \cos^2 \theta - 1) + 3$$

$$= \cos 4\theta + 4(\cos 2\theta) + 3$$

$$\therefore \cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$$

$$7(b) i) LHS = \cos(k\theta - \theta) - 2\cos\theta \cos k\theta$$

$$= \cos k\theta \cos \theta + \sin k\theta \sin \theta - 2\cos\theta \cos k\theta$$

$$= \sin k\theta \sin \theta - \cos\theta \cos k\theta$$

$$RHS = -\cos(k\theta + \theta)$$

$$= -\cos k\theta \cos \theta + \sin k\theta \sin \theta$$

$$\therefore LHS = RHS \quad (1)$$

(ii) Step 1 Let $n=1$

$$LHS = 1, RHS = \frac{1 - \cos\theta - \cos\theta + \cos\theta}{2 - 2\cos\theta}$$

$$= \frac{2 - 2\cos\theta}{2 - 2\cos\theta} = 1$$

Step 2 $\therefore k=1$ satis proposition.

Suppose prop. true for $n=k$, integer

i.e. suppose $1 + \dots + \cos(k-1)\theta = \frac{1 - \cos\theta - \cos k\theta + \cos(k-1)\theta}{2 - 2\cos\theta}$

Step 3 Now $1 + \dots + \cos k\theta = \frac{1 - \cos\theta - \cos k\theta + \cos(k-1)\theta}{2 - 2\cos\theta} + \cos k\theta$

$$= \frac{1 - \cos\theta - \cos k\theta + \cos(k-1)\theta + 2\cos k\theta - 2\cos k\theta}{2 - 2\cos\theta}$$

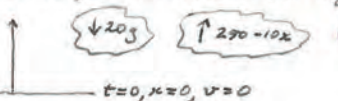
$$= \frac{1 - \cos\theta + \cos k\theta - \cos(k-1)\theta}{2 - 2\cos\theta} \text{ (using (i))}$$

$$= \frac{1 - \cos\theta + \cos((k+1)\theta) - \cos k\theta}{2 - 2\cos\theta}$$

$\therefore$ prop true when $n=k+1$

Step 4 If true for $n=1$ then true for $n=1+1=2$ etc

As true for $n=1$, $\therefore$ true for all pos integers n

(c) 

$$F = ma$$

$$\therefore 20a = 250 - 10x - 20g$$

$$\therefore a = \frac{250}{20} - \frac{1}{2}x$$

$$\therefore v \cdot \frac{dv}{dx} = \frac{25}{2} - \frac{1}{2}x$$

$$\therefore \int v \cdot dv = \int \left(\frac{25}{2} - \frac{1}{2}x\right) dx$$

$$\therefore \left[\frac{1}{2}v^2\right]_0^v = \left[\frac{25}{2}x - \frac{1}{4}x^2\right]_0^x$$

$$\therefore \frac{1}{2}v^2 = 25x - \frac{1}{4}x^2$$

$$\therefore v = 0, \text{ stationary} \quad (3)$$

7 (a) Let $P(x) = (x-1)(x-2)Q(x) + ax + b$

Let $x=1 \therefore P(1) = a + b$

But $P(x) = (x-1)Q(x) + 3 \Rightarrow P(1) = 3$

$$\therefore a + b = 3 \quad (1)$$

Similarly, let $x=2, 5 = 2a + b \quad (2)$

$$(2) - (1) \therefore a = 2, b = 1$$

$$\therefore \text{remainder } 2x + 1 \quad (2)$$

(k) (i) $LHS = \frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta - i\cos\theta} \times \frac{1 + \sin\theta + i\cos\theta}{1 + \sin\theta + i\cos\theta}$

$$= \frac{1 + 2\sin\theta + \sin^2\theta - \cos^2\theta + 2i\cos\theta + 2i\sin\theta}{1 + 2\sin\theta + \sin^2\theta + \cos^2\theta}$$

$$= \frac{2\sin\theta + 2\sin^2\theta + (1 + \sin\theta)2i\cos\theta}{2(1 + \sin\theta)}$$

$$= \sin\theta + i\cos\theta = RHS \quad (3)$$

(ii) $LHS = (\sin\theta + i\cos\theta)^n$

$$= [\cos(\frac{\pi}{2} - \theta) + i\sin(\frac{\pi}{2} - \theta)]^n$$

$$= \cos^n(\frac{\pi}{2} - \theta) + i\sin^n(\frac{\pi}{2} - \theta)$$

Using de Moivre's Theorem

$$= RHS \quad (2)$$

8. (a) (i) Let $\angle ADC = \alpha$

In $\triangle ADC$, $\angle PAD = \angle DCP = 90^\circ$ (angle bet. tangent & radius is 90°)

$\therefore PADC$ is a cyclic quad. (opp. angles add to 180°)

In $\triangle BDC$, $\angle DCQ = \angle DBQ = 90^\circ$ (similarly)

$\therefore CDPQ$ is a cyclic quad (similarly)

(ii) $\angle ADC = \angle BDC = \alpha$ (ext. angle of $\triangle$)

cyclic quad $DBQC$ equal to remote int. $\angle$

$AD = DC$ (equal tangent) $\angle BQ = \angle C$ (equal radii)

$\therefore \triangle ADC$ & $\triangle BDC$ are isosceles

(2 pairs of equal sides)

Also, $\angle ADC = \angle BDC = \alpha$

$\therefore \triangle ADC \parallel \triangle BDC$ (2 equal angles)

sides about the angles in the same ratio

(iii) $\angle QCB = 180 - \alpha$ using angle sum in $\triangle$ & 2 equal angles

Also $\angle APC = 180 - \alpha$ (opp. angles supp. in cyclic quad)

$\therefore \angle DPC = 180 - \alpha$ (kites $ADCP$ has diag. bisect $\angle$)

$\therefore PD \parallel CB$ (corresp. $\angle$ s equal)

OR using $ADCP$ is a cyclic quad, one can show that $\angle DPC = \angle CAD = \angle QCB$

$\therefore PD \parallel CB$ (corresp. $\angle$ s equal)

8 (b) $P \sin \frac{\theta}{2} = \sin \frac{\theta}{2} + 2\cos\theta \sin \frac{\theta}{2} + 2\cos 2\theta \sin \frac{\theta}{2} + 2\cos 3\theta \sin \frac{\theta}{2}$

(i) $= \sin \frac{\theta}{2} + \sin(\theta + \frac{\theta}{2}) - \sin(\theta - \frac{\theta}{2}) + \sin(2\theta + \frac{\theta}{2}) - \sin(2\theta - \frac{\theta}{2})$

$$+ \sin(3\theta + \frac{\theta}{2}) - \sin(3\theta - \frac{\theta}{2})$$

$$= \sin \frac{\theta}{2} + \sin \frac{3\theta}{2} - \sin \frac{\theta}{2} + \sin \frac{5\theta}{2} - \sin \frac{3\theta}{2} + \sin \frac{7\theta}{2} - \sin \frac{5\theta}{2}$$

$$= \sin \frac{7\theta}{2} \quad (2)$$

(ii) Let $\theta = \frac{2\pi}{7} \therefore P \sin \frac{2\pi}{14} = \sin \pi$

$$\therefore P = 0$$

$$\therefore 1 + 2\cos\theta + 2\cos 2\theta + 2\cos 3\theta = 0 \quad (2)$$

(iii) $\cos 2\theta = 2\cos^2\theta - 1$ & $\cos 3\theta = \cos(2\theta + \theta)$

$$= \cos 2\theta \cos \theta - \sin 2\theta \sin \theta$$

$$= (2\cos^2\theta - 1)\cos\theta - 2\sin\theta \cos\theta \sin\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta \sin^2\theta$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta(1 - \cos^2\theta)$$

$$= 2\cos^3\theta - \cos\theta - 2\cos\theta + 2\cos^3\theta$$

$$= 4\cos^3\theta - 3\cos\theta$$

$$\therefore P = 1 + 2\cos\theta + 2(2\cos^2\theta - 1) + 2(4\cos^3\theta - 3\cos\theta)$$

$$= 1 + 2\cos\theta + 4\cos^2\theta - 2 + 8\cos^3\theta - 6\cos\theta$$

$$= -1 - 4\cos\theta + 4\cos^2\theta + 8\cos^3\theta$$

$$= 8x^3 + 4x^2 - 4x - 1 \text{ where } x = \cos\theta$$

But $P=0$ when $\theta = \frac{2\pi}{7} \therefore x = \cos \frac{2\pi}{7}$ & $8x^3 + 4x^2 - 4x - 1 = 0$

$$(3)$$